

Milestone 1 – Summary of Background Study and Source Identification

Sensing Leeds Air Quality: Predictive Monitoring and Instrument Performance

MATH5872M – Dissertation in Data Science and Analytics

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1. Understanding of the Topic

Leeds runs the largest deployment of low-cost PM_{2.5} sensors in the UK outside London, set up jointly by the University and the City Council. Each sensor reports every two minutes, so the data has both a dense spatial spread across the city and a very fine time resolution. The contrast with the regulatory Defra network (the Automatic Urban and Rural Network, or AURN) is what makes this interesting. The AURN relies on a handful of accurate reference instruments that report hourly averages. An hourly average smooths away anything brief, so a short, sharp pollution episode lasting a few minutes barely shows up. The two-minute Leeds data does not have that problem: it can pick out exactly these short-lived, spatially confined features.

The plan is to use this data to build what the brief calls a “standard view” of the city. In practice that means a baseline description of how each location normally behaves, together with a picture of how the locations relate to one another. Once you know what normal looks like, two practical questions become answerable. Is a sensor currently reading well above its own norm, which would point to a real local pollution event worth reporting to the Council? Or has a sensor stopped reporting properly, which instead suggests a fault and a site visit? The two cases are different in kind. One is about the air; the other is about the instrument. The catch is that both can look like “unusual data” at first glance, so telling them apart is the real difficulty.

Missing and unreliable data sits underneath all of this. Low-cost devices drop offline, return values that make no physical sense, and respond to temperature and humidity, so whatever method I use has to cope with gaps rather than assume a complete record. That is also why the brief asks specifically for classification and clustering methods that work with missing data, illustrated on both simulated and real datasets, and ultimately for a tool that flags exceedances to the Council and identifies sensors that need checking.

2. Methods in the Suggested Literature

The two papers in the brief come from the same University College Cork group and supply most of the method. Byrne et al. (2023) used a 16-sensor PurpleAir network in Cork and showed that PM_{2.5} varies a lot even within one city, with daily peaks in some areas roughly twice those elsewhere, mostly because of residential solid-fuel burning. Working at the two-minute scale, they were able to separate brief local spikes (a nearby chimney, say) from the slower regional background, finding that some sensors were affected by local sources 40–50% of the time and others under 20%. The takeaway for Leeds is that the high-resolution data carries a genuine local signal that hourly averaging would lose.

Byrne et al. (2024) build on this with the concentration similarity index (CSI), the recent method the brief points to. The CSI is a single number summarizing how alike two sensors’ concentration profiles are over a chosen period. Computing it for every pair of sensors gives a CSI matrix that describes the spatial structure of the whole network. They then cluster that matrix using hierarchical clustering and fuzzy c-means, checking the result with silhouette and Calinski–Harabasz scores. In both Cork and Dungarvan this pulled out two location types: busier central or residential sites with more solid-fuel pollution, and cleaner sites on the edge of the urban area. One finding stood out: similarity followed location type more than physical distance. They also showed that when a sensor sits next to a reference instrument, the CSI can say how representative that regulatory site is of the wider city. For Leeds, the CSI matrix is a sensible way to construct the standard view, and the co-location idea maps neatly onto the AURN sites in the city.

3. Additional Source Beyond the Brief

The CSI papers describe the network and its structure well, but they say little about the detection rule itself: how you actually decide a single new reading is unusual, and whether that reading is a real event or a sensor fault. To fill that gap I found **van Zoest, Stein and Hoek (2018)**, “**Outlier Detection in Urban Air Quality Sensor Networks**”, which addresses exactly this for a 35-sensor network in Eindhoven.

Their approach is to classify before thresholding. Instead of judging every reading against one citywide cutoff, they split a year of data into 16 classes defined by site type (urban background or traffic), day type (weekday or weekend) and time of day, then model the values within each class with a truncated normal distribution. A reading counts as an outlier only if it is far from the mean of its own class given that class’s spread. In Eindhoven this flagged just

0.1–0.5% of readings while keeping the real spatial and temporal variation intact, which is the point: a blunt global threshold would have buried genuine urban variation under false alarms.

Three things make the paper useful here. It draws the exact distinction the brief needs, between *errors* (faults, mishandling, poor siting) and *events* (genuinely high or low concentrations), which is the line between “visit the sensor” and “alert the Council.” It also offers a testable way to tell them apart: a real event tends to last a while and show up across several neighboring sensors at once, whereas an isolated one-sensor blip looks more like a fault. And the authors discuss running the method in near real time with a moving window, which is the setting the Leeds tool needs. It complements the CSI work rather than repeating it. The CSI fixes what normal looks like across the network; van Zoest et al. supply the logic for acting on departures from it.

4. Commentary and Reflection

So far the reading has given me a clear sense of the aims and a reasonable grasp of how the CSI works, though I would not yet say I understand the methods in depth. I can follow the CSI as defined in the 2024 paper and the reasoning behind the adaptations Byrne et al. make to it, and van Zoest et al. was helpful for the separate question of how to flag an unusual reading and tell an event from a fault.

The part I am least sure about is missing data, which is central to the brief but not really covered in the CSI papers, so I plan to read more on classification and clustering when data are incomplete. There are also some practical details about the Leeds data I do not know yet, such as how the sensors are arranged, whether any are co-located with AURN sites, and how gaps and faulty readings are recorded, which I expect to become clear once I have the dataset.

Before I have the real data, the most useful next steps seem to be working carefully through the CSI derivation and reproducing it on a small example, and trying the outlier method from van Zoest et al. on a simple test case. Since simulated data is already available from my supervisors, I can also use it to check the methods against a known answer before moving to the real network, rather than having to generate test data myself.

5. Schematic Plan

- **Literature and methods (now – Meeting 1):** consolidate the CSI pipeline (Byrne et al. 2023, 2024) and the outlier framework (van Zoest et al. 2018); read further on clustering with missing data.
- **Data familiarisation:** once the Leeds dataset is available, profile coverage, gaps, units, sensor metadata and any AURN co-location.
- **Method testing on simulated data (Meetings 1–2):** use the simulated data provided by the supervisors to reproduce the CSI and trial the outlier method against a known answer.
- **Standard view (by Meeting 2):** compute the CSI matrix on the real data and cluster locations to set the baseline “normal” structure.
- **Detection and classification (toward Meeting 3 / Milestone 2):** implement exceedance flagging and an event-versus-fault rule using persistence and spatial spread; test on real data.
- **Tool and evaluation (Meetings 3–5):** prototype a near-real-time reporting tool and assess how well it performs.
- **Drafting and write-up (Meeting 6 / Milestone 3, then final submission):** assemble the results into the dissertation and refine ahead of the final deadline.
- **Final dissertation:** due 28 August.

References

1. Byrne, R., Ryan, K., Venables, D. S., Wenger, J. C., and Hellebust, S. (2023). Highly local sources and large spatial variations in PM_{2.5} across a city: evidence from a city-wide sensor network in Cork, Ireland. *Environmental Science: Atmospheres*, 3, 919–930. <https://doi.org/10.1039/d2ea00177b>
2. Byrne, R., Wenger, J. C., and Hellebust, S. (2024). Spatial analysis of PM_{2.5} using a concentration similarity index applied to air quality sensor networks. *Atmospheric Measurement Techniques*, 17, 5129–5146. <https://doi.org/10.5194/amt-17-5129-2024>
3. van Zoest, V. M., Stein, A., and Hoek, G. (2018). Outlier detection in urban air quality sensor networks. *Water, Air, & Soil Pollution*, 229, 111. <https://doi.org/10.1007/s11270-018-3756-7>